$a \equiv b \pmod{n}$ iff $\mathrm{rem}(a,n) = \mathrm{rem}(b,n)$

iff $a = b + \underset{\uparrow \text{int}}{k}\, n$

exmp: prove that the relation $\equiv \pmod 3$ is an equivalence relation on $\mathbb{Z}$.

- reflexive $\forall x \in \mathbb{Z}: \; x \equiv x \pmod 3$

$$x = x + \underset{\uparrow \text{int}}{0} \times 3 \Rightarrow x \equiv x \pmod 3 \checkmark$$

- Symmetry $\forall x,y \in \mathbb{Z}: \; x \equiv y \pmod 3 \Rightarrow y \equiv x \pmod 3$

$$x = y + \underset{\uparrow \text{int}}{k} \times 3$$

$$y = x - k \times 3$$

$$y = x + \underset{\uparrow \text{int}}{(-k)} \times 3 \text{ therefore } y \equiv x \pmod 3 \checkmark$$

- transidivity $\forall x,y,z \in \mathbb{Z}: \; x \equiv y \pmod 3 \wedge y \equiv z \pmod 3 \Rightarrow x \equiv z \pmod 3$

(1) $x = y + \underset{\uparrow \text{int}}{k} \times 3$

$y = z + \underset{\uparrow \text{int}}{k'} \times 3$ (2)

(1) $x = y + k \times 3 \overset{(2)}{=} z + k' \times 3 + k \times 3 = z + \underbrace{(k'+k)}_{\text{int}} \times 3$

$x = z + \text{int} \times 3 \Rightarrow x \equiv z \pmod 3 \checkmark$

$\equiv \pmod 3$ is equivalence relation on $\mathbb{Z}$

Partial order (poset): A relation that is reflexive, antisymmetric, and transitive.

reflexive
symmetry — equivalence
partial order — antisymmetry
transition

exmp: Is $\leqq$ on $A = \{2, 3, 4, 5\}$ a partial order?

- reflexive : $\forall_{x \in A} : x \leqq x$ ✓
- transitive: $\forall_{x,y,z \in A} : x \leqq y \land y \leqq z \Rightarrow x \leqq z$ ✓
- antisymmetric: $\forall_{x,y \in A} : x \leqq y \land y \leqq x \Rightarrow x = y$ ✓

$(\leqq, A)$ is a poset

$\leqq$ is a partial order on $\mathbb{Z}$: $(\leqq, \mathbb{Z})$ is a poset

$(\geqq, \mathbb{Z})$ is a poset.

$(<, \mathbb{Z})$ is not a poset because $<$ is not reflexive.

exmp: $A = \{1, 2, 3, 4\}$

$R_1 = \{(1,1), (1,2), (1,4), (2,4), (3,3), (4,4)\}$

reflexive ✗ transitive ✓ antisymmetry ✓ Not partial order

$R_2 = \{(1,1), (2,2), (2,3), (2,4), (3,2), (3,3), (3,4), (4,4)\}$

reflexive ✓ transitive ✓ antisymmetric ✗ Not partial order

$x | y$: x divides y (y is divisible by x)

int   int

$y = k \times x$

some int

$4|8$ $3|15$ $3|3$

$15 = 5 \times 3$ $3 = 1 \times 3$

int   int

exmp: Is $(|, \mathbb{N})$ a poset? $\mathbb{N} = \mathbb{Z}^+$

reflexive: $\forall x \in \mathbb{N} : x | x$ ✓ → YES

$x = 1 \times x$

int

antisymmetry: $\forall x, y \in \mathbb{N} : x|y \wedge y|x \Rightarrow x = y$ ✓

$y = k \times x$ $x = k' \times y$

int ≥ 0 int ≥ 0

$y = k \times x = k \times k' \times y$ because $y, x > 0$ since $y, x \in \mathbb{N}$

$\Rightarrow k \times k' = 1$ → this is only possible if $k = 1$ and $k' = 1$

$y = k \times x = 1 \times x \Rightarrow y = x$ ✓   k and k' cannot be -1

transitive: $\forall_{x, y, z \in \mathbb{N}} : x|y \wedge y|z \Rightarrow x|z$

$x|y : y = k \times x$ k: int

$y|z : z = k' \times y$ k': int

$z = k' \times y = k' \times k \times x = \underbrace{(k' \times k)}_{\text{int}} \times x$ thus $x|z$ ✓

$\Rightarrow (|, \mathbb{N})$ is a poset.

exmp: Is $(|, \mathbb{Z})$ a poset? No

$5 \mid -5$ $\quad$ $-5 \mid 5$ $\quad$ but $5 \neq -5$

$-5 = (-1) \times 5$
↑
int

$5 = (-1) \times (-5)$
↑
int

therefor $(|, \mathbb{Z})$ is not antisymmetric

exmp: $A = \{1,2,3\}$. Is $\subseteq$ a partial order on $2^A = P(A)$.

$\{3,2\}$

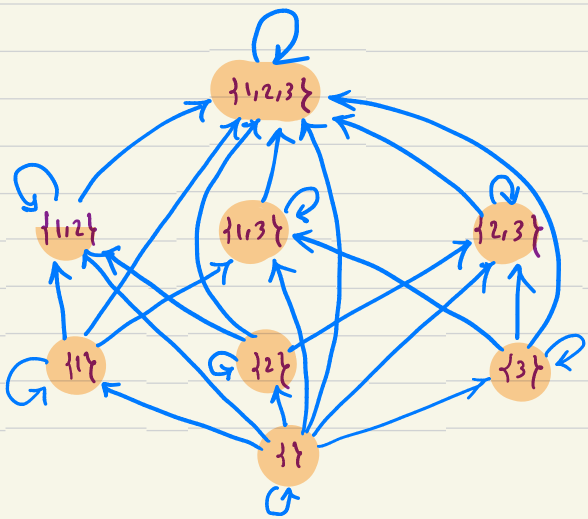
$2^A = \{\{\}, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$

reflexive: $\forall x \in 2^A : x \subseteq x$ ✓

antisymmetry: $\forall x,y \in 2^A : x \subseteq y \wedge y \subseteq x \Rightarrow x = y$ ✓

transitivity: $\forall x,y,z \in 2^A : x \subseteq y \wedge y \subseteq z \Rightarrow x \subseteq z$ ✓

$(\subseteq, 2^A)$ is a poset



questions: let R and S be two partial orders on a set A.

Determine if $R \cap S$ and $R \cup S$ are partial orders on A?

$R \cap S$: our relation

reflexive: $\forall_{x \in A}$: $(x,x) \in R \cap S$ ✓

$$\left.\begin{array}{l} R \text{ is reflexive} : (x,x) \in R \\ S \text{ is reflexive} : (x,x) \in S \end{array}\right\} (x,x) \in R \cap S$$

transitive: $\forall_{x,y,z}$: $(x,y) \in R \cap S \wedge (y,z) \in R \cap S \Rightarrow (x,z) \in R \cap S$

$$\left.\begin{array}{l} (x,y) \in R \quad \wedge \quad (y,z) \in R \xrightarrow{R \text{ transitive}} (x,z) \in R \\ (x,y) \in S \quad \wedge \quad (y,z) \in S \xrightarrow{S \text{ transitive}} (x,z) \in S \end{array}\right\} (x,z) \in R \cap S \ ✓$$